Day-24 Pandas

🔹 1. What is Pandas?

- Pandas is a **Python library** used for:
  - Data analysis
  - Data manipulation

✅ Key Points:

- Simplifies complex data operations
- Works with structured data (tables)
- Widely used in **Data Science & AI**

🔹 2. Importance of Pandas

🚀 Why Pandas is Important:

- Handles **large datasets efficiently**
- Provides **Excel-like functionality**
- Helps in:
  - Data cleaning
  - Data transformation

- Data analysis

📌 Industry Fact:

- Used in **~80% of data science workflows**

🔹 3. Core Data Structures

📌 1. Series

- One-dimensional data structure
- Example:

```
[10,20,30]
```

📌 2. DataFrame

- Two-dimensional (rows × columns)
- Most commonly used
- Like a table in Excel

✅ Features:

- Labeled data
- Indexed rows & columns

🔹 4. Creating a DataFrame

📌 Example:

```
importpandasaspd

data= {'Name': ['Alice','Bob'],'Age': [25,30]}
df=pd.DataFrame(data)

print(df)
```

🧠 Concept:

- Dictionary keys → Column names
- Values → Row data

🔹 5. Reading Data from Files

📌 Functions:

```
df=pd.read_csv('file.csv')
df=pd.read_excel('file.xlsx')
df=pd.read_json('file.json')
```

✅ Purpose:

- Load real-world data into Pandas

🔹 6. Data Viewing Functions

📌 Preview Functions:

```
df.head()# First 5 rows
df.tail()# Last 5 rows
```

📌 Summary Functions:

```
df.info()# Structure of data
df.describe()# Statistical summary
```

🔹 7. Data Selection

📌 Selecting Columns:

```
df['Age']
```

📌 Selecting Rows:

```
df.loc[0]# By label
df.iloc[0]# By index
df.loc[0,'Name']# Specific value
```

🧠 Tip:

- `loc` → label-based
- `iloc` → position-based

🔹 8. Data Cleaning

📌 Handling Missing Data:

```
df.dropna()# Remove missing values
df.fillna(0)# Replace with 0
df.fillna('N/A')# Replace with text
df.isna()# Detect missing values
```

✅ Importance:

- Improves accuracy of analysis

🔹 9. Data Operations

📌 Basic Functions:

```
df['Age'].mean()# Average
df['Score'].max()# Maximum
df['Price'].min()# Minimum
```

🧠 Use:

- Quick statistical insights

🔹 10. Filtering Data

📌 Example:

```
df[df['Age']>18]
```

✅ Concept:

- Uses conditions
- Returns only matching rows

🔹 11. Real-Life Applications

📌 Where Pandas is Used:

- Student data analysis
- Business reporting
- Machine learning preprocessing

📊 Industry Insight:

- Used in most **data science projects**

🔹 12. Advantages of Pandas

- Easy to learn
- Saves time
- Powerful and flexible
- Handles large data efficiently

🔹 13. Conclusion / Key Takeaways

🎯 Pandas Helps You:

- Simplify data analysis
- Handle large datasets
- Work efficiently with Python